Tractor Tyre Pressure & Ballast Guide

Metric Standards (bar / kPa) for Soil Preparation, General Utility & Water Ballast Capacity

Key Pressure Principles: Soil Prep vs. General Work

- **Soil Preparation (Heavy Tillage):** Lower pressures increase tire contact patch, maximizing traction, reducing wheel slip to 10–15%, and preventing subsoil compaction.
- **General Work & Road Travel:** Higher pressures reduce tread wear, rolling resistance, and sidewall flex when carrying heavy mounted implements or trailers.
- **Radial vs. Bias-Ply:** Radial tires can run safely at much lower pressures (0.8–1.2 bar) in the field compared to Bias-Ply tires (1.2–1.8 bar).

1. FRONT TYRE PRESSURE RECOMMENDATIONS (METRIC)

Tyre Size	Tyre Type	Soil Prep / Field Work	General Work & Road
6.00-16 / 7.50-16	Bias-Ply (2WD)	1.6 – 1.8 bar (160–180 kPa)	2.2 – 2.5 bar (220–250 kPa)
280/85R24 (11.2R24)	Radial (4WD)	0.8 – 1.0 bar (80–100 kPa)	1.3 – 1.6 bar (130–160 kPa)
320/85R24 (12.4R24)	Radial (4WD)	0.8 – 1.0 bar (80–100 kPa)	1.3 – 1.6 bar (130–160 kPa)
340/85R24 (13.6R24)	Radial (4WD)	0.9 – 1.1 bar (90–110 kPa)	1.4 – 1.7 bar (140–170 kPa)
380/85R24 (14.9R24)	Radial (4WD)	0.9 – 1.1 bar (90–110 kPa)	1.4 – 1.7 bar (140–170 kPa)
420/70R24	Wide Radial (4WD)	0.8 – 1.0 bar (80–100 kPa)	1.2 – 1.5 bar (120–150 kPa)
380/85R28 (14.9R28)	Radial (4WD)	1.0 – 1.2 bar (100–120 kPa)	1.5 – 1.8 bar (150–180 kPa)
420/85R28 (16.9R28)	Radial (4WD)	1.0 – 1.2 bar (100–120 kPa)	1.5 – 1.8 bar (150–180 kPa)
Front Loader Operating	Any (Heavy Load)	Add +0.4 to +0.6 bar over standard setting to protect sidewalls	

2. REAR TYRE PRESSURE RECOMMENDATIONS (METRIC)

3. LIQUID BALLAST (WATER FILLING) GUIDE

Standard Liquid Filling Rule: 75% Capacity (Valve Stem Level)

To add traction weight without creating a harsh ride or risking rim damage, fill tires to **75% capacity**. Rotate the tire so the valve stem is at the 12 o'clock (top) position; fill with water until liquid starts coming out of the valve port, leaving 25% air cushion at the top.

Note: Plain water weighs exactly 1.0 kg per liter (1000 L = 1000 kg).

4. WATER BALLAST VOLUME & ADDED WEIGHT CHART (PER TYRE)

Tyre Size	Tyre Position / Class	Water Fill Volume (Liters)	Added Weight (kg)
9.5-16	Small Front (4WD)	45 L	+45 kg
11.2-24 / 280/85R24	Medium Front (4WD)	90 L	+90 kg
12.4-24 / 320/85R24	Medium Front (4WD)	115 L	+115 kg
13.6-24 / 340/85R24	Large Front (4WD)	145 L	+145 kg
14.9-24 / 380/85R24	Front / Utility Rear	180 L	+180 kg
12.4-28	Standard Utility Rear	130 L	+130 kg
13.6-28 / 340/85R28	Standard Utility Rear	165 L	+165 kg
14.9-28 / 380/85R28	Mid-Range Rear	200 L	+200 kg
16.9-28 / 420/85R28	Mid-Range Rear	260 L	+260 kg
16.9-30 / 420/85R30	Heavy Rear	315 L	+315 kg
18.4-34 / 460/85R34	Heavy Rear	415 L	+415 kg
18.4-38 / 460/85R38	Broadacre Rear	450 L	+450 kg
20.8-38 / 520/85R38	High-Horsepower Rear	530 L	+530 kg

Anti-Freeze Note for Cold Climates: If operating in freezing temperatures, mix Calcium Chloride (CaCl_2) or non-toxic antifreeze (Bio-Glycol) into the water. A standard mixture of $0.4\text{ kg } \text{CaCl}_2$ per liter of water prevents freezing down to -12°C and increases ballast density to approx. 1.2 kg per liter.